



CSCI 236: Data Structures and Algorithms (Fall 2025)

Class time: Monday/Wednesday/Friday, 10:00 – 11:05am (In-Person)

Location: Paine 212

Instructor	Babafemi Sorinolu ○ Email: babafemi.sorinolu@houghton.edu ○ Office: Paine 209B; 585.567.9259 ○ Office hours: Tuesday/Thursday, 2:30-4:30pm (Feel free to email me to schedule a virtual meeting.)
Moodle Page	https://moodle.houghton.edu/course/view.php?id=29601
Course Description	(4 credit hours) - This class covers the fundamental data structures of computer science and accompanying algorithms. Linked lists, stacks, queues, binary trees, red-black trees, heaps and other abstract data types are included. Classical sorting and searching algorithms are presented and implemented. Prerequisite: NA.
Optional Textbooks	Robert Lafore , Data Structures and Algorithms in Java 2nd Edition ISBN10: 0672324539 Mark Allen Weiss , Data Structures and Problem-Solving Using Java 4th Edition ISBN10: 0321541405

Long-Range Goals:

- Understand the key properties (structure and methods) of several data structures including lists, stacks, queues, trees, priority queues, heaps, hash tables and graphs.
- Create efficient programs using the data structures covered in the course.
- Implement and use standard sorting and searching algorithms.
- Develop traversal algorithms for trees and graphs, and identify shortest paths and minimum spanning trees in graphs.
- Strengthen and expand understanding of programming in Python and/or Java.

Course Objectives: After successful completion of this course, students will be able to:

- Understand fundamental data structures and select those that effectively model the information in a given problem.
- Evaluate efficiency trade-offs among different data structure implementations or combinations.
- Apply algorithm analysis techniques to assess the performance of algorithms and compare data structures.
- Implement a variety of data structures in detail, along with standard searching and sorting algorithms, and apply them appropriately.
- Recognize common design patterns and determine when their use improves software design.
- Apply object-oriented design principles to data structures in medium-scale software systems.

Activities & Time Commitment: For a four-credit course, you should expect to commit at least 12 hours per week. Although not every week will be the same, these are typical weekly times:

- Class (3 hours): These would take the form of presentation-based instruction and in-class activity.
- Reading, study, & review (3 hours): There will be readings assigned each week. The student is expected to both read the text and take notes. This activity (along with the class presentation) will form the basis of the three exams that will be given during the course.
- Homework (3 hours): Along with the readings, there will be some assignments based on the assigned readings and class presentations. These assignments are designed to reinforce the understanding of key concepts in computer architecture through various exercises. Work is due at the time it is assigned, whether you are in class or not.
- Activities and Project (3 hours): Each student will do a variety of hands-on activities in and out of the class lecture times. These will vary from simple coding to a student-selected project using Python programming language. This project will require the transference of the concepts and skills of the course to an area of application of interest to the student.

Evaluation and Grading: The grading and grading scale will be based on the following tables:

- **Grading:** The evaluation will be based on the following criteria and weights.

Criteria	Weights (%)
Homework	30
Class Hands-on Activities/Quizzes	30
Midterm Exam	20
Final Exam/Project	20

- **Grading Scale**

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
%	100-93	92-90	89-87	86-83	82-80	79-77	76-73	72-70	69-67	66-63	62-60	59-

All final grades will be rounded up to the nearest whole number when assigning letter grades.

Attendance: Let me know (by email to babafemi.sorinolu@houghton.edu with the subject "CSCI 236 Attendance") if and why you will be missing a class. In general, this should be done before the class you will miss.

Exams: There will be one midterm exam in the class and one cumulative final exam/project. Make-up exams will only be given in extenuating circumstances; prior arrangements must be made if possible.

Tentative dates for each exam are as follows:

- Midterm Exam: Monday, October 6th, 10:00–11:05am
- Final Exam: Monday, December 15th, 10:30am-12:30pm.

Late Submission Policy: Assignments are due at the stated deadline. Late submissions will be accepted for up to **one week with a 20% deduction**. After one week, the assignment will receive no credit unless prior arrangements have been made or there are documented extenuating circumstances. **Exams, quizzes, and in-class activities cannot be submitted late.**

Academic Honesty: Honesty is the foundation on which all intellectual endeavors rest. To use the ideas of others without acknowledging the authors of those ideas belies the nature and purpose of academic life. At Houghton, where we strive to live out Christian calling and commitment, personal integrity, including academic honesty, should be the hallmark of all

our work and relationships. Students are expected to exhibit extreme care relative to personal honesty in all academic work, including in-class and out-of-class learning experiences, such as exams, quizzes, journals, papers, and research projects. Please refer to the catalog for the details of this policy ([Link](#)).

AI Policy: AI tools (e.g., ChatGPT, GitHub Copilot, Gemini) may be used only as a supplement for learning—such as clarifying algorithm concepts, exploring problem-solving strategies, or checking syntax. They may not be used to directly generate solutions to assignments, projects, or exams. **All submitted work must reflect your own problem-solving process and understanding.** If AI is used in preparation, you must note it briefly (e.g., “Consulted ChatGPT for explanation of recursion”). Misuse (e.g., submitting AI-generated solutions or code you cannot explain) will be treated as an academic integrity violation.

Self-Reporting of Disabilities: If you have an academic or physical disability that requires accommodations, it is up to you to self-report any such disability to the office of Academic Support and Accessibility Services in the Center for Student Success located on the 1st floor of Chamberlain. (585-567-9622) With appropriate documentation, you will be afforded the necessary accommodation. For more information about the Center for Academic Success and Advising, go to: <https://www.houghton.edu/current-students/center-for-student-success/academic-support-and-accessibility-services/>

Tentative Schedule: Below is a tentative calendar for the topics we will cover during each class period. These topics correspond to the chapters in our textbook.

Week	Class Date	Lecture Topics	Chapters
1	9/1 - 9/5	Introduction to Data Structures	1
2	9/8 - 9/12	Introduction to complexity: O ()-notation	-
3	9/15 - 9/19	Arrays	2
4	9/22 - 9/26	Searching and Sorting Algorithms	3, 7
5	9/29 - 10/3	Stacks	4
6	10/6 - 10/10	Queue	4
7	10/13 - 10/17	Linked Lists	5
8	10/24	Recursion and Advanced Sorting	6
9	10/27 - 10/31	Trees	8
10	11/3 - 11/7	Hash Tables	11
11	11/10 - 11/14	Heaps	12
12	11/17 - 11/21	Graphs	13
13	11/24	Applying Data Structure I	15
14	12/1 - 12/5	Applying Data Structure II	15
15	12/8 - 12/10	Review	
16	12/15	Final Exams	